PERSONAL INFORMATION

Hai Jiang

Guangdong Province, P. R. China

(+86) 198 6771 3757

jianghai97@mail.nankai.edu.cn

https://pigejianghai.github.io/Hai.github.io

https://github.com/pigejianghai

https://www.linkedin.com/in/hai-jiang-649171314

ORCID 0009-0007-9735-7581

Gender Male | Date of birth 21 December 1997 | Nationality Chinese

JOB APPLIED FOR

Ph.D. Applicant (Medical Image Analysis / Computer Vision)

WORK EXPERIENCE

July 2026–September 2026

Algorithm Development Engineer

Shenzhen Guardian Glow Technology Co., Ltd.

- Developed eye-tracking algorithms on image-sensor data for AR/VR glasses, covering wearing-state recognition and gaze-direction estimation with data analysis, feature construction, and model training/evaluation.
- Fused CNN visual features and eye geometric features, and used PCA to analyze wearing-offset effects; contributed to model lightweighting and embedded inference optimization.
- Performed feature engineering, dimensionality reduction, and classifier tuning; deployed and optimized models on embedded targets.

Skills Eye tracking, gaze estimation, PCA, CNN-based regression, model lightweighting, embedded deployment.

April 2026–July 2026

Image Algorithm Engineer

Shenzhen Image Technology Co., Ltd.

- Develop C++ geometry algorithms for PCB copper-layer Gerber parsing and polygon feature extraction, converting raw contour data into structured inspection-ready representations.
- Extract edges, traces, corners, blocks, pads, and other manufacturing-critical geometric entities from large-scale polygon sets for downstream process inspection and quality analysis.
- Design computational geometry and topology analysis modules, optimize memory usage and throughput, and maintain unit tests and algorithm documentation for production use.

Skills C++, computational geometry, topology analysis, Gerber parsing, unit testing, technical documentation.

August 2025–April 2026

Computer Vision Algorithm Engineer

Shenzhen Zhongrui Power Technology Co., Ltd.

- Built an automated AI engine for a property-management data platform using YOLO, enabling object detection, instance segmentation, and classification of buildings and facilities from user-uploaded images.
- Integrated OCR for text extraction from buildings, units, and signs through an end-to-end pipeline covering preprocessing, localization, recognition, and structured output.
- Packaged the inference pipeline behind standardized APIs for web integration, using ONNX export and TensorRT acceleration to meet real-time response requirements.

Skills YOLO, OCR, ONNX, TensorRT, API integration, computer vision deployment.

July 2022–July 2025 **Research Assistant**

Computational Medical Imaging Laboratory, Sun Yat-sen University

- Designed and validated algorithms for national medical AI projects covering MRI, dual-energy CT, and digital breast tomosynthesis.
- Proposed an anatomy-guided multitask learning framework with attention for placenta accreta spectrum classification on MRI, achieving AUC $>$ 0.99 and first-author acceptance at ISBI 2025.
- Built a multi-channel CNN with a virtual-class strategy for lymph-node metastasis prediction from dual-energy CT, improving the average AUC to 0.85 (under revision at Medical Physics).
- Contributed to multi-view CNN and Siamese-network research for DBT architectural-distortion detection, supporting a Guangdong Science and Technology Progress Award; also contributed to grant proposals, final reports, and industry collaboration.

Skills Medical imaging, deep learning, PyTorch, TensorFlow, data preprocessing, research writing.

EDUCATION

September 2019–June 2022 **Master of Science in Computational Mathematics** M. Sc.

Nankai University (NKU, Project 985 & 211, Double First-Class), Tianjin, China

Thesis Deep Unfolding Networks for MRI Image Reconstruction.

Research on optimization algorithms (Primal-Dual Learning, ADMM), deep learning, and MRI reconstruction.

Key Skills Deep unfolding, ADMM-based optimization, PyTorch, C++, Python.

Supervisors Prof. Yunhua Xue, Prof. Chunlin Wu.

Relevant Coursework Approximation Theory, Numerical Optimization, Functional Analysis, Matrix Computation, Numerical PDEs.

GPA 77.4/100.0.

September 2015–June 2019 **Bachelor of Engineering in Information Security** B. Eng.

Lanzhou University (LZU, Project 985 & 211, Double First-Class), Lanzhou, Gansu, China

Thesis Improved Upper Bounds of Roman Domination Number in Maximal Outerplanar Graphs.

Focused on graph theory and combinatorial optimization.

Supervisor Prof. Zepeng Li.

Relevant Coursework Discrete Mathematics, Data Structures, Algorithm Design and Analysis, Machine Learning, Cryptography, C/C++ Programming, Database Theory.

GPA 4.15/5.00.

RESEARCH EXPERIENCE

July 2022–July 2025 **Computational Medical Imaging Laboratory** R. A.

School of Computer Science and Engineering, Sun Yat-sen University

Conducted three medical-imaging research projects spanning MRI-based placenta accreta spectrum classification, DECT-based metastasis prediction, and DBT multi-view detection.

Project 1 Placenta Accreta Spectrum Disorder Classification

- Developed an anatomy-guided multitask learning framework with attention for PASD binary classification, achieving AUC $>$ 0.99 (20% improvement over baseline).
- Built a multi-task learning model using T2-WI MRI images for subtype classification, achieving AUC of 0.80.
- First-author paper accepted at ISBI 2025: “Anatomy-Guided Multitask Learning for MRI-Based Classification of Placenta Accreta Spectrum and Its Subtypes” [1].

Skills Literature review, data preprocessing, PyTorch, research writing.

Project 2 Breast Cancer Metastasis Prediction

- Designed a multi-channel CNN with a virtual-class strategy on dual-energy CT scans across 11 energy levels (40–140 keV) to predict metastasis.
- Achieved average AUC of 0.85 (10% gain over baseline).
- First-author manuscript under revision at Medical Physics [4].

Skills TensorFlow, Keras, data analysis, experimental design, research writing.

Project 3 Multi-view Learning for Atypical Architectural Disorder Detection in Breast DBT

- Contributed to the multi-view computer-aided detection model with ipsilateral learning for atypical architectural disorder detection in digital breast tomosynthesis (DBT) images.
- Introduced a novel Siamese module with triplet loss to enhance feature extraction and cross-view detection consistency.
- Co-authored paper published in Physics in Medicine & Biology (2023) [2]; research supported a Guangdong Science and Technology Progress Award.

Skills Mask-RCNN, Detection, Siamese Network, Multi-view, Triplet-Loss.

January 2022–June 2022

Image Analysis Team

School of Mathematical Sciences, Nankai University

Project Deep ADMM-Net Implementation for Compressive-Sensing MRI.

- Implemented an end-to-end deep unfolding reconstruction network based on the ADMM optimization framework for compressive-sensing MRI.
- Improved reconstruction quality under undersampling using PyTorch and C++, grounded in compressed sensing and ADMM analysis.

Skills Compressive-sensing theory, deep unfolding, ADMM optimization, MRI reconstruction.

PUBLICATIONS

[1] **Hai Jiang**, Qiongting Liu, Yuanpin Zhou, Jiawei Pan, Ting Song, and Yao Lu. "Anatomy-Guided Multitask Learning for MRI-Based Classification of Placenta Accreta Spectrum and its Subtypes". In: *2025 IEEE 22nd International Symposium on Biomedical Imaging (ISBI)*. 2025, pp. 1–5.

[2] Jiawei Pan, Zilong He, Yue Li, Weixiong Zeng, Yaya Guo, Lixuan Jia, **Hai Jiang**, Weiguo Chen, and Yao Lu. "Atypical architectural distortion detection in digital breast tomosynthesis: a multi-view computer-aided detection model with ipsilateral learning". In: *Physics in Medicine & Biology* 68.23 (2023), p. 235006.

[3] XueFang Wang, **Hai Jiang**, Ruxu Du, Yong Zhong, and Yao Lu. "Anatomical-Prior-Based Multiscale Segmentation of Cardiac Substructures Using Enhanced Skip-Connections and a Triple-View Fusion Network". Submitted.

[4] **Hai Jiang**, Chushan Zheng, Jiawei Pan, Yuanpin Zhou, Qiongting Liu, Xiang Zhang, Jun Shen, and Yao Lu. *DECT-based Space-Squeeze Method for Multi-Class Classification of Metastatic Lymph Nodes in Breast Cancer*. 2025. arXiv: 2505.17528 [eess.IV]. URL: https://arxiv.org/abs/2505.17528.

[5] Jiawei Pan, Zilong He, Yue Li, Weixiong Zeng, Yaya Guo, Lixuan Jia, **Hai Jiang**, Weiguo Chen, and Yao Lu. "Multi-view Architectural Distortion Detection with Confidence Boosting in Digital Breast Tomosynthesis". Under revision. 2025.

TECHNICAL SKILLS

Programming Python, C/C++, PyTorch, TensorFlow, Keras, OpenCV, MATLAB, LaTeX, Git.

Tools & Platforms Linux (Ubuntu), ONNX, TensorRT, CMake, Microsoft Office, Adobe Photoshop.

Research Methods Deep learning (CNNs, YOLO, nnU-Net), computer vision, image processing, medical imaging, computational geometry, compressive sensing, eye tracking.

AWARDS & SCHOLARSHIP

2015–2019 Four-time recipient of the Third-Class Merit Scholarship for Academic Excellence at LZU.

2019–2022 Three-time recipient of the Third-Class Merit Scholarship for Academic Excellence at NKU.

LANGUAGE PROFICIENCY

Mandarin Native.

English Professional (C1; IELTS 7.0, CET6 476/710, CET4 544/710).

Cantonese Intermediate.

RESEARCH PROJECT DOCUMENTATION

Funding Proposal Writing — Contributed to National Key R&D Program of China [No. 2023YFE0204300].

Report Writing — Contributed to completion and progress reports for National Natural Science Foundation of China [No. 81971691, 12126610] and R&D Program of Pazhou Lab [No. 2023K0606].

TEACHING EXPERIENCE

Courses Taught — Calculus, Mathematical Analysis.

Thesis Supervision — Breast Cancer Classification Method Based on Dual-Energy CT Images.

RESEARCH INTERESTS

Artificial Intelligence, Deep Learning (CNNs, GNNs).

Mathematics, Medical Image Analysis, Physics.

REFERENCES

Yunhua Xue — Associate Professor, Computational Mathematics, Nankai University
E-mail: yhxue@nankai.edu.cn
Phone: (+86)022-23506423
Address: 94 Weijin Road, Nankai District, Tianjin, China.

Yao Lu — Professor, Medical Image Analysis, Sun Yat-sen University
E-mail: luyao23@mail.sysu.edu.cn
Phone: (+86)18664509325
Address: Xingang West Road, Haizhu District, Guangzhou, China.

Yuanpin Zhou — Research Scientist, Machine Learning, Shanghai Runda Medical Technology Co., Ltd.
E-mail: yuanpinzhou@gmail.com
Phone: (+86)15521227642
Address: Hangzhou, Zhejiang, China.